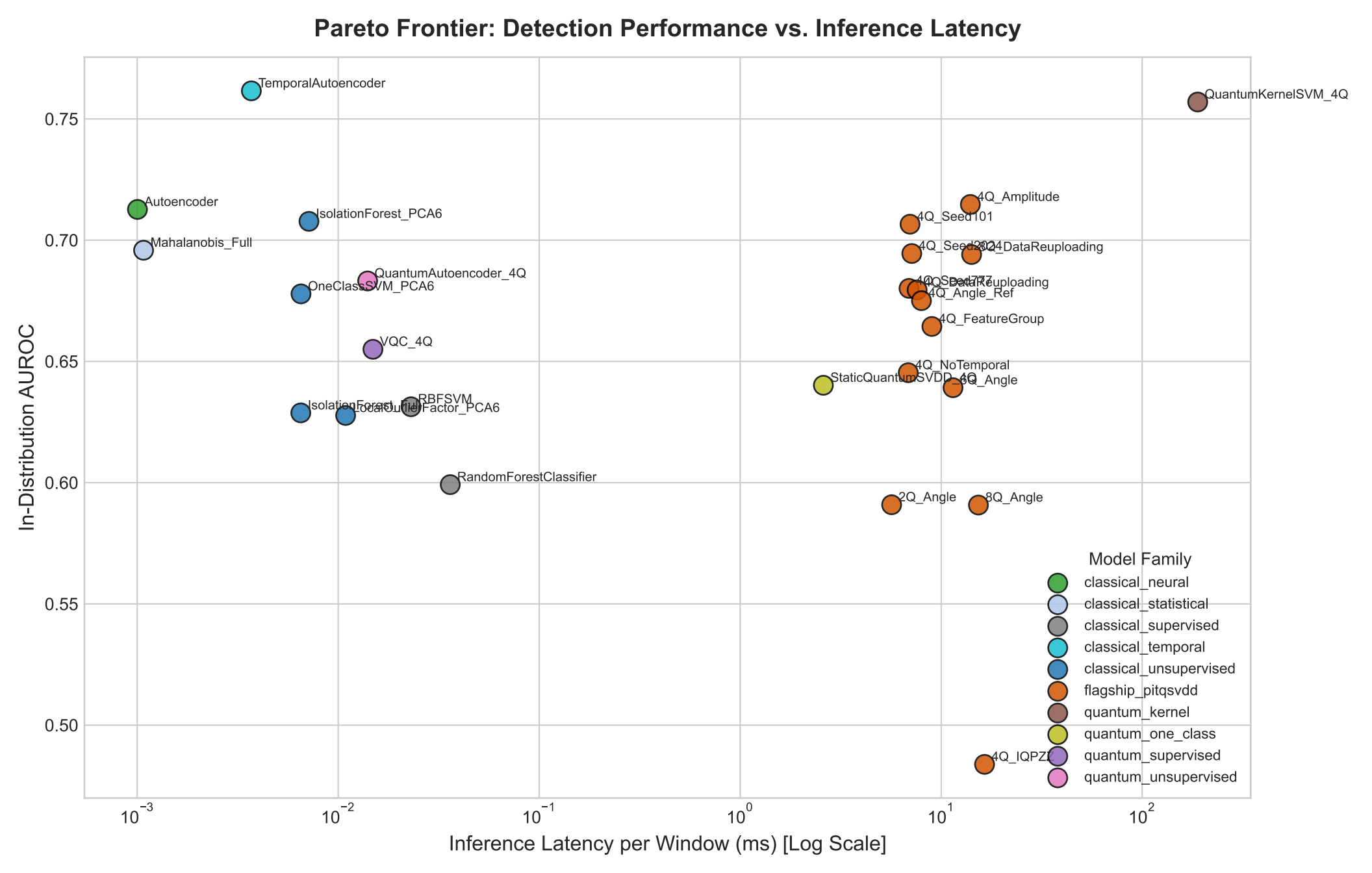
Pareto Frontier: Detection Performance vs. Inference Latency
In-Distribution AUROC
Inference Latency per Window (ms) [Log Scale]
0.75
0.70
0.65
0.60
0.55
0.50
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
TemporalAutoencoder
QuantumKernelSVM_4Q
Autoencoder
IsolationForest_PCA6
Mahalanobis_Full
QuantumAutoencoder_4Q
OneClassSVM_PCA6
VQC_4Q
RBFSVM
LocalOutlierFactor_PCA6
RandomForestClassifier
4Q_Amplitude
4Q_Seed101
4Q_DataReuploading
4Q_Angle_Ref
4Q_FeatureGroup
4Q_NoTemporal
StaticQuantumSVDD_4Q
2Q_Angle
8Q_Angle
Model Family
classical_neural
classical_statistical
classical_supervised
classical_temporal
classical_unsupervised
flagship_pitqsvdd
quantum_kernel
quantum_one_class
quantum_supervised
quantum_unsupervised